

11.30 Solution: Following Eq. (11.149), $\mathbf{D}$ and $\mathbf{H}$ should transform similarly, with

$$\mathbf{D} = \gamma (\mathbf{D}' - \boldsymbol{\beta} \times \mathbf{H}') - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{D}'), \quad \mathbf{H} = \gamma (\mathbf{H}' + \boldsymbol{\beta} \times \mathbf{D}') - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{H}').$$

Using the constitution relations $\mathbf{D}' = \varepsilon \mathbf{E}'$ and $\mathbf{H}' = \mathbf{B}'/\mu$, we have

$$\mathbf{D} = \gamma \left(\varepsilon \mathbf{E}' - \frac{1}{\mu} \boldsymbol{\beta} \times \mathbf{B}' \right) - \frac{\varepsilon \gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}'), \quad \mathbf{H} = \gamma \left(\frac{\mathbf{B}'}{\mu} + \varepsilon \boldsymbol{\beta} \times \mathbf{E}' \right) - \frac{\gamma^2}{\mu(\gamma + 1)} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{B}').$$

Now, applying Eq. (11.149) for $\mathbf{E}$ and $\mathbf{B}$, $\mathbf{D}$ becomes

$$\begin{aligned} \mathbf{D} &= \gamma \left\{ \varepsilon \left[\gamma (\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \right] \right. \\ &\quad \left. - \frac{1}{\mu} \boldsymbol{\beta} \times \left[\gamma (\mathbf{B} - \boldsymbol{\beta} \times \mathbf{E}) - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{B}) \right] \right\} \\ &\quad - \frac{\varepsilon \gamma^2}{\gamma + 1} \boldsymbol{\beta} \left(\boldsymbol{\beta} \cdot \left[\gamma (\mathbf{E} + \boldsymbol{\beta} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \right] \right) \\ &= \gamma^2 \varepsilon \mathbf{E} + \gamma^2 \varepsilon \boldsymbol{\beta} \times \mathbf{B} - \frac{\varepsilon \gamma^3}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \\ &\quad - \frac{\gamma^2}{\mu} \boldsymbol{\beta} \times \mathbf{B} + \frac{\gamma^2}{\mu} \boldsymbol{\beta} \times (\boldsymbol{\beta} \times \mathbf{E}) + \frac{\gamma^3}{\mu(\gamma + 1)} (\boldsymbol{\beta} \times \boldsymbol{\beta}) (\boldsymbol{\beta} \cdot \mathbf{B}) \\ &\quad - \frac{\varepsilon \gamma^3}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) - \frac{\varepsilon \gamma^3}{\gamma + 1} \boldsymbol{\beta} [\boldsymbol{\beta} \cdot (\boldsymbol{\beta} \times \mathbf{B})] + \frac{\varepsilon \gamma^4}{(\gamma + 1)^2} \beta^2 \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \\ &= \gamma^2 \varepsilon \mathbf{E} + \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) (\boldsymbol{\beta} \times \mathbf{B}) + \frac{\gamma^2}{\mu} \boldsymbol{\beta} \times (\boldsymbol{\beta} \times \mathbf{E}) - \frac{2\varepsilon \gamma^3}{\gamma + 1} \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) + \frac{\varepsilon \gamma^4}{(\gamma + 1)^2} \beta^2 \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}), \end{aligned}$$

since $\boldsymbol{\beta} \times \boldsymbol{\beta} = 0$ and $\boldsymbol{\beta} \cdot (\boldsymbol{\beta} \times \mathbf{B}) = 0$. Notice that

$$\frac{\gamma^4 \beta^2}{(\gamma + 1)^2} - \frac{2\gamma^3}{\gamma + 1} = \frac{\gamma^2(\gamma - 1) - 2\gamma^3}{\gamma + 1} = -\gamma^2,$$

then

$$\begin{aligned} \mathbf{D} &= \gamma^2 \varepsilon \mathbf{E} + \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) (\boldsymbol{\beta} \times \mathbf{B}) + \frac{\gamma^2}{\mu} \boldsymbol{\beta} \times (\boldsymbol{\beta} \times \mathbf{E}) - \varepsilon \gamma^2 \boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{E}) \\ &= \gamma^2 \varepsilon \mathbf{E} + \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) (\boldsymbol{\beta} \times \mathbf{B}) + \frac{\gamma^2}{\mu} \boldsymbol{\beta} \times (\boldsymbol{\beta} \times \mathbf{E}) - \varepsilon \gamma^2 [\beta^2 \mathbf{E} + \boldsymbol{\beta} \times (\boldsymbol{\beta} \times \mathbf{E})] \\ &= \gamma^2 (1 - \beta^2) \varepsilon \mathbf{E} + \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) (\boldsymbol{\beta} \times \mathbf{B}) - \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) [\boldsymbol{\beta} \times (\boldsymbol{\beta} \times \mathbf{E})] \\ &= \varepsilon \mathbf{E} + \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) (\beta^2 \mathbf{E}_\perp + \boldsymbol{\beta} \times \mathbf{B}), \end{aligned}$$

where we have used the fact that $\gamma^2 = (1 - \beta^2)^{-1}$ and $\boldsymbol{\beta} \times (\boldsymbol{\beta} \times \mathbf{E}) = \beta^2 \mathbf{E}_\perp$.

The result for $\mathbf{H}$ can be obtained in a similar way and will be omitted here.